

03. AM Squad: Tech Stack & Tooling Overview

 Page is about?

***Purpose:** This page provides a **universal setup guide** for **SDET/Automation engineers** in the GS Automation team. It outlines the complete tooling landscape used across **Prime, Astro, Jett, Unite, and Universal Enrollment** branches. Ideal for **onboarding new hires**, it ensures a **step-by-step walkthrough** of core tooling, environment configuration, IDE setup, and execution prerequisites.*

Section 1 – Core Tech Stack Overview

Category	Tool(s)	Purpose / Notes
Language	Java (17+)	Primary scripting language; set JAVA_HOME
Build Tool	Apache Ant / Maven (v3.9.3+)	Dependency management and build lifecycle
Framework	TestNG, Cucumber	Test execution and BDD scripting
Web Driver	Selenium 3.141 / 4.6+	Browser automation; v4.6+ supports auto-driver mgmt
IDE	IntelliJ / Eclipse	IDEs for developing and debugging automation code
Execution	Jenkins	CI/CD integration; remote suite execution
Versioning	GitLab	Automation Repo
Reporting	TestNG Console / Surefire Plugin	Results via target/surefire-reports/ + logs
Script Utils	Groovy (optional)	Used in specific Prime resource modules

Section 2 – Local Environment Setup (Step-by-Step)

Step 1: Copy Core Tool Folders

Source:
L:\Departments\Financial Services\UII\UII\UII-QA\Cucumber_Automation\automation-env
Recommended Destination: C:\

Required Folders:

- java
- ant
- maven (apache-maven-3.9.3+)
- chromedriver (e.g., 135.0.7049.95)
- cucumber
- groovy
- testng
- selenium-3.141.59 and selenium-java-4.1.0
- startSeleniumGrid.bat
- GitLab Cert (for HTTPS repo access)

Step 2: Configure System Environment Variables

Variable	Sample Value
JAVA_HOME	C:\Program Files\Java\jdk-17
ANT_HOME	C:\ant\apache-ant-1.9.4
MAVEN_HOME	C:\Program Files\apache-maven-3.9.9
CUCUMBER_HOME	C:\cucumber\4.2.0
SELENIUM_HOME	C:\selenium\selenium-3.141.59
TESTNG_HOME	C:\testng\testng-6.9.9
GROOVY_HOME	C:\groovy\groovy-2.4.11
PRIME_HOME	C:\Workspace\Gitlab\Automation\unite-test-automation\unite

%JAVA_HOME%\bin;%ANT_HOME%\bin;%MAVEN_HOME%\bin;%SELENIUM_HOME%;%PRIME_HOME%

Run set in CMD to verify

Restart **Eclipse**. **IntelliJ**.

Restart **Eclipse**, **IntelliJ**, or the **PC** if variables are not applied immediately.

```
A:\Users\vtalasil>set
ALLUSERSPROFILE=C:\ProgramData
ANT_HOME=C:\Sant'apache-ant-1.9.4
APPDATA=C:\Users\vtalasil\AppData\Local\Roaming
CLIENT_DEV_PATH=C:\MSI-Clients\1
CLIENT_DEV_PATH_MSI_PORT=13075
CLIENT_DEV_PATH_MSI_HOST=8
CLIENT_DEV_PATH_MSI_PORT=18880
CommonProgramFiles=C:\Program Files\Common Files
CommonProgramFiles(x86)=C:\Program Files (x86)\Common Files
CommonProgramW6432=C:\Program Files\Common Files
COMPTEL=C:\Program Files\Lenovo\Communications Utility
COMPUTERNAME=M490
SYSTEMDRIVE=C:\Windows\system32\cmd.exe
CNUMBER_HOME=C:\cnumber\2.3.1
IP_NO_HOST_CHECK=NO
GROOVY_HOME=C:\groovy\groovy-2.4.11
HOMEDRIVE=C:
HOMEPATH=C:\Users\vtalasil
JAVA_HOME=C:\Program Files\Java\jdk1.8.0_151
LOCALAPPDATA=C:\Users\vtalasil\AppData\Local
LOGONSERVERS=[*]NTLM - PWDRES02
NUMBER_OF_PROCESSORS=4
OS=Windows_NT
SystemRoot=C:\Windows\System32\bin\x86\bin\%SystemRoot%\system32\bin\x86\bin\netlm\sechelp\sechelp.dll
SYSTEM32\WindowsPowerShell\v1.0\cmd.exe
TEMP=C:\Users\vtalasil\AppData\Local\Temp
TMPDIR=C:\Users\vtalasil\AppData\Local\Temp
TUT=C:\Program Files (x86)\lenovo
USERHOME=C:\Users\vtalasil\My Documents
USERDOMAIN=ntlm
USERPROFILE=C:\Users\vtalasil
windir=C:\Windows
windows_tracing_flags=3
windows_tracing_logfile=C:\BVTBin\Tests\installpackage\csilogfile.log
C:\Users\vtalasil>
```

IntelliJ IDEA (Preferred)

- Install from official website or shared drive
- Enable plugins:
 - **Cucumber for Java**
 - **TestNG**
 - **Groovy** (if required)

Setup Guide: [IntelliJ – Setup & GitLab Repo Management](#)

- Unzip from shared drive:
L:\Departments\Financial Services\UII\UII\Software\eclipse-jee-<version>.zip
- Import projects:
File > Import > Maven > Existing Maven Projects
- Set encoding: UTF-8 (Project Properties Resource)
- Install Cucumber Plugin via Marketplace
- Right-click project Configure Convert to Cucumber Project

Setup Guide: [Eclipse – Setup & IDE Configuration](#)

Setup Guide: [GitLab – Setup & Project Cloning](#)

Section 3 – Project & Framework Structure

Folder/Module	Purpose
prime-core	Base execution and Java logic (TestNG, utilities)
prime-cucumber	Step definitions and runners
prime-lib	Project-specific configs (unite, astro, etc.)
qa-resource-*	Data loaders, SQL scripts, env configs
samples/	Example POMs, tests, configs

docs/	Prime framework documentation
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Section 4 – Execution & Profiles (Maven)

Task	Command Example
Build resource modules	cd qa-resource mvn clean install
Build Prime framework	cd prime-parent mvn clean install
Generate documentation	mvn clean generate-resources
Run default profile	mvn clean install
Run custom profile	mvn clean install -P<profile-id>
Override test suite	-Dxml.suite=path/to/testng.xml
Debug via Eclipse	-Dmaven.surefire.debug (port 5005)

Section 5 – Reporting & Logging

Type	Location
TestNG Report	target/surefire-reports/
Log File	logs/execution.log

Use JVM flags:

```
-Dlogging.console=DEBUG
-Dlogging.file=DEBUG
```

Section 6 – Additional Tools & Utilities

Tool	Notes / Location
Postman	API testing – L:\Departments\Financial Services\UII\UII\UII-F0DEV\software\postman
Notepad++	Text editor – L:\Departments\Financial Services\UII\UII\UII-F0DEV\installers\notepad++
7-Zip	Compression – L:\Departments\Financial Services\UII\UII\Software\7Zip
Oracle Client	Required for SQL Developer Verify DB access Configure tnsnames.ora
Cygwin	CLI utilities – C:\apps\cygwin64
Perforce	Legacy – No longer required for Git-based repo
IntelliJ/Eclipse	Follow setup above • Copy L:\Departments\Financial Services\UII\UII\Software\intelliJ\idealU-<version>.exe to your c:\temp and then install • Copy L:\Departments\Financial Services\UII\UII\Software\ eclipse-jee-<version>-SR2-win32-x86_64.zip to your c:\temp and then extract to c:\ ◦ Eclipse Challenges.docx • ⚠️ TODO: Create new pages and link back to here, for: ◦ GS 529 Onboarding Use Eclipse P4V plugin
Jenkins	Ask DevOps for job configs & Access
Oracle SQL developer	SQL client - L:\Departments\Financial Services\UII\UII\UII-B0DEV

Section 7 – Environment Access & Network Tools

- [Environment Access & Setup](#)

- [General Environment Setup – Putty, DB Tunnels, SQL Developer](#)

Section 8 – Tools & Software:

All required tools can be downloaded from the file directory below. Download to your C:\temp folder and then run/extract from there.

Location:

- [L:\Departments\Financial Services\UI\UI\UI-FODEV\software](#)
- [L:\Departments\Financial Services\UI\UI\UI-BODEV\Software_NewLaptop](#)
- [L:\Departments\Financial Services\UI\UI\Software](#)

Final Setup Checklist

- All tools copied and installed
- Environment variables configured
- Maven, JDK, GitLab setup completed
- IntelliJ or Eclipse setup with plugins
- QA-Resource and Prime modules built
- mvn clean install executed successfully
- DB & VPN access validated